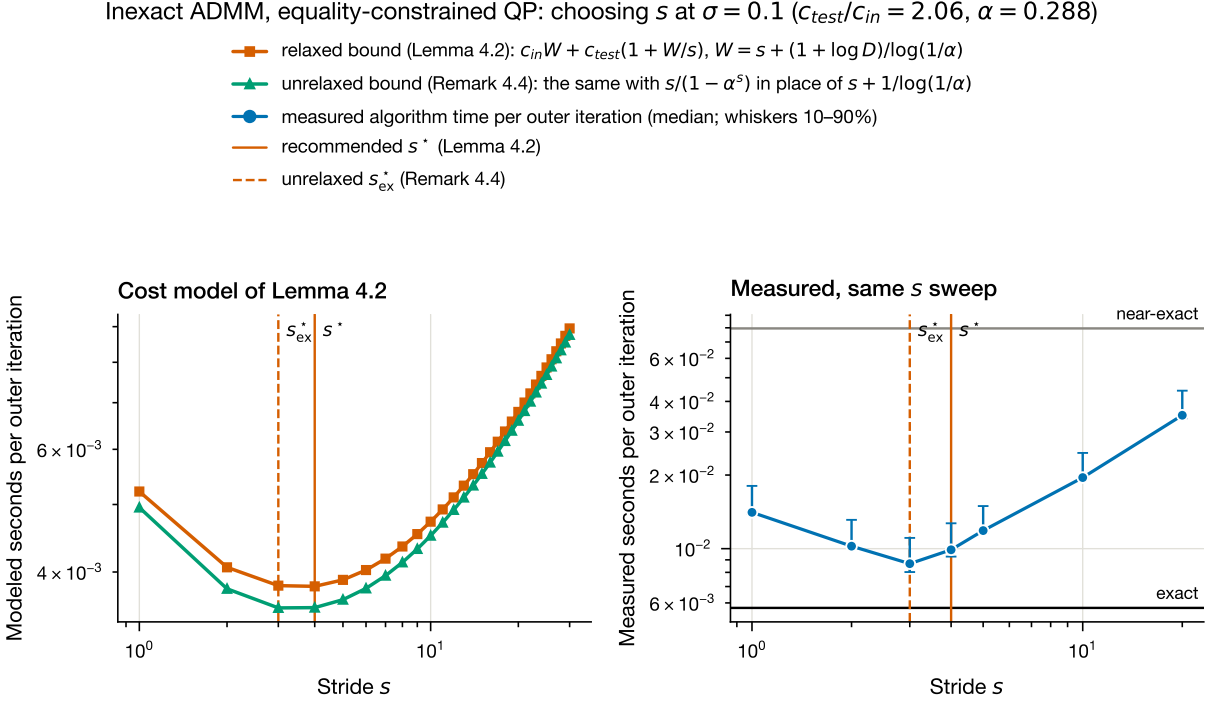
Inexact ADMM, equality-constrained QP: choosing s at $\sigma = 0.1$ ($c_{test}/c_{in} = 2.06$, $\alpha = 0.288$)
relaxed bound (Lemma 4.2): $c_{in}W + c_{test}(1 + W/s)$, $W = s + (1 + \log D)/\log(1/\alpha)$
unrelaxed bound (Remark 4.4): the same with $s/(1 - \alpha^s)$ in place of $s + 1/\log(1/\alpha)$
measured algorithm time per outer iteration (median; whiskers 10–90%)
recommended $s^\star$ (Lemma 4.2)
unrelaxed $s^\star_{ex}$ (Remark 4.4)
Cost model of Lemma 4.2
Modeled seconds per outer iteration
6×10^{-3}
4×10^{-3}
$s^\star_{ex}$
$s^\star$
10^0
10^1
Stride s
Measured, same s sweep
Measured seconds per outer iteration
near-exact
exact
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
$s^\star_{ex}$
$s^\star$
10^0
10^1
Stride s